10x10x8 Stasis Chamber: Master Build Plan & Testing Protocol

This document serves as the official structural and testing protocol for the 10x10x8 Faraday tent, designed for high-resolution, dual-subject EEG hyperscanning (the OpenBCI Mark IV setup).

Hardware Store Cut & Material List

Category	Item Description	Quantity	Notes
PVC Framing	1.5-inch Schedule 40 PVC Pipe (10-foot lengths)	17	Use for bottom/top perimeters, verticals, and roof cross-brace.

Category	Item Description	Quantity	Notes
PVC Joints	1.5-inch Furniture-Grade Fittings	20x Tees, 4x 4-Way, 5x T-Fittings, 1x 4-Way Cross	Avoid standard shallow plumbing hubs.
Adhesives & Ties	PVC Primer & Cement, Heavy-Duty Zip-Ties	1 set each	Purple primer and standard cement.
Faraday Mesh	Woven Copper Mesh (36" or 48" wide rolls)	520+ sq ft total	Do NOT use 6-inch strips.
Seam Sealant	2-inch Conductive Copper Foil Tape	2+ Rolls	Must have conductive adhesive.
Grounding	8-foot Copper Grounding Rod & Brass Clamp & Bonding Wire	1	Drive into earth outside the barn.
Dielectric Barrier	Concrete Curing Blankets (Vinyl/Insulated)	Sufficient for 10x10 area	Isolates copper floor from concrete rebar.

Phase 1: Ground & Foundation

- **The Earth Ground:** Drive the 8-foot copper rod into the dirt outside the barn. Run the solid copper wire inside, leaving it coiled near the build site.
- **The Dielectric Pad:** Lay the concrete curing blankets in the corner (strict 5-foot gap from metal walls). Tape the seams completely flat.

Phase 2: The Skeletal Frame (1.5" PVC)

1. **The Floor:** Assemble the 10x10 bottom square using 3-way corners. Prime and cement it.
2. **The Floor Mesh:** Lay the bottom sheet(s) of 36" or 48" copper mesh over the PVC square. Use the Fold-and-Tape Seam Protocol to join the floor strips. Wrap the outer edges under the PVC.
3. **The Verticals:** Insert and cement the four 8-foot vertical pipes.
4. **The Roof:** Build the top 10x10 square with the 4-way corners and center T-fittings.

Connect the center 4-way cross-brace to prevent sagging. Drop the roof onto the verticals and cement every joint.

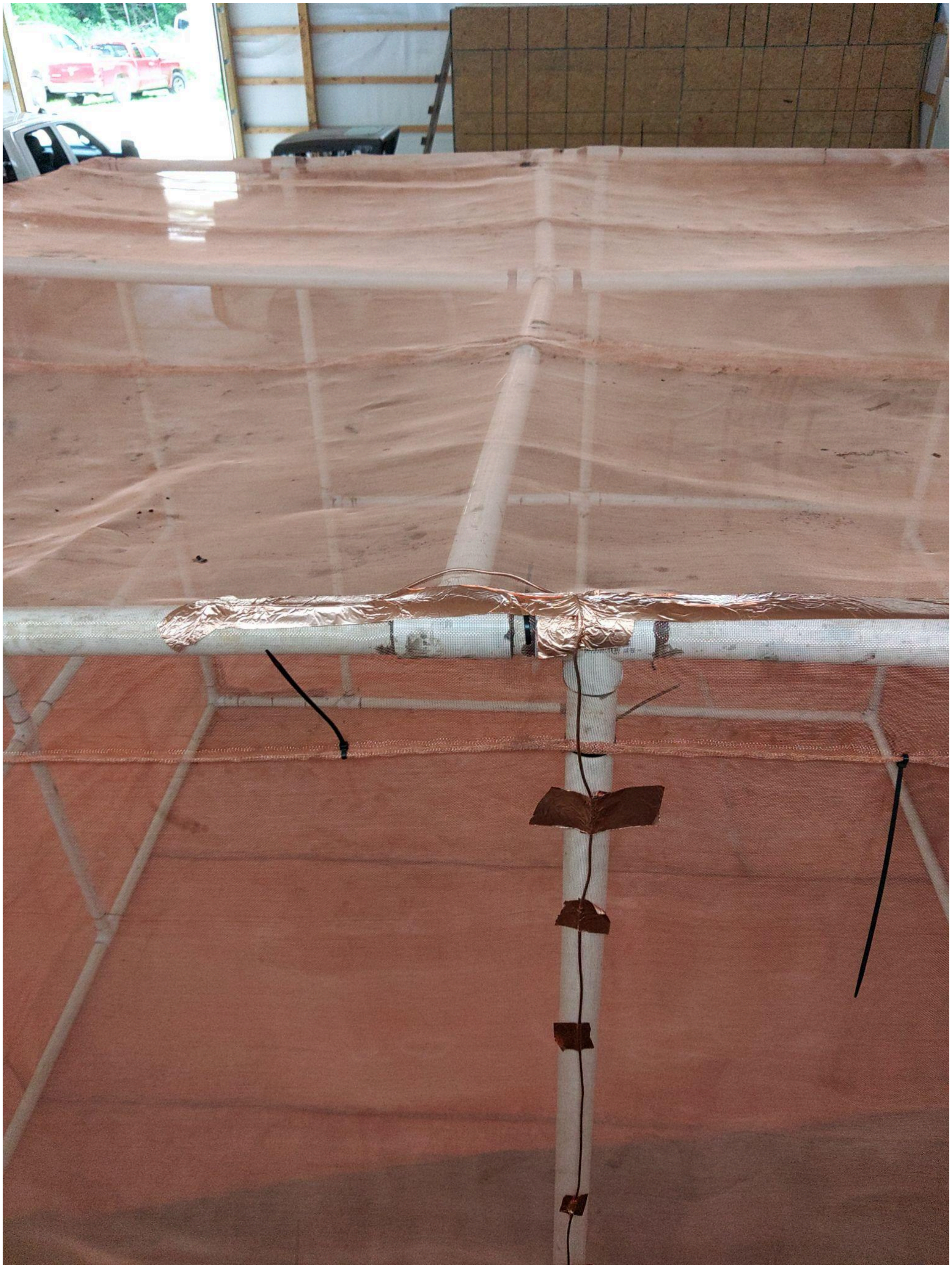
Phase 3: The Envelope & The Seam Protocol

Unroll your wide mesh vertically, creating panels that hang from the top PVC bar down to the floor. Zip-tie them to the frame. When joining two vertical panels, execute this exact sequence:

- **Overlap:** Overlap the two mesh panels by exactly 2 inches.
- **Fold:** Fold that 2-inch overlap back on itself by 1 inch (like a hem).
- **Crush:** Use a hard rubber roller to crush the fold completely flat, locking the copper threads together.
- **Tape:** Run your 2-inch conductive copper foil tape straight down the center of the crushed fold.
- **Bond to Floor:** Use copper tape to bond the bottom edge of the wall panels directly to the floor mesh.

Phase 4: The Door & The Ground

1. **The Flap:** Zip-tie a massive, oversized mesh panel to the top front PVC bar. Build the magnetic hem at the bottom and sides so it snaps tight to the walls and floor.
2. **The Grounding Bond:** Run a bare copper wire around the lower perimeter of the tent, taping it securely to the mesh. Clamp your exterior Earth Ground wire directly to this loop.



Phase 5: Proof of Concept Testing Protocol

Execute these three tests to verify the structural and electromagnetic integrity of the stasis chamber before initiating human trials.

Test 1: The AM Radio Hack (Low-Frequency Test)

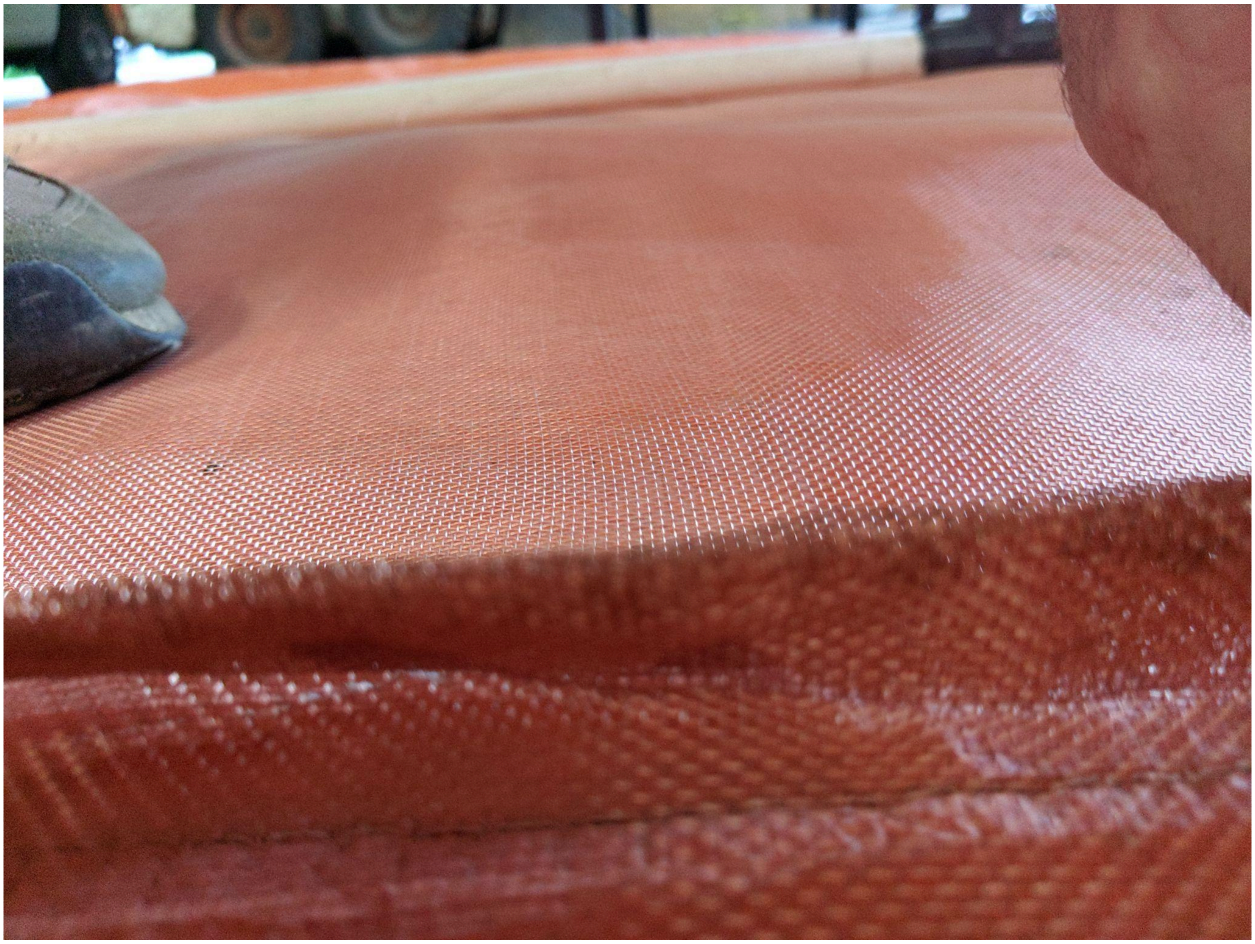
- **Method:** Tune a cheap, battery-powered portable AM/FM radio to a dead AM station so it is blasting pure static.
- **Execution:** Walk inside the tent with the radio and drop the magnetic flap to seal the entrance.
- **Success Criteria:** The static must instantly choke out and go completely silent. If you can still hear the static clearly, there is a low-frequency leak in one of your seams.

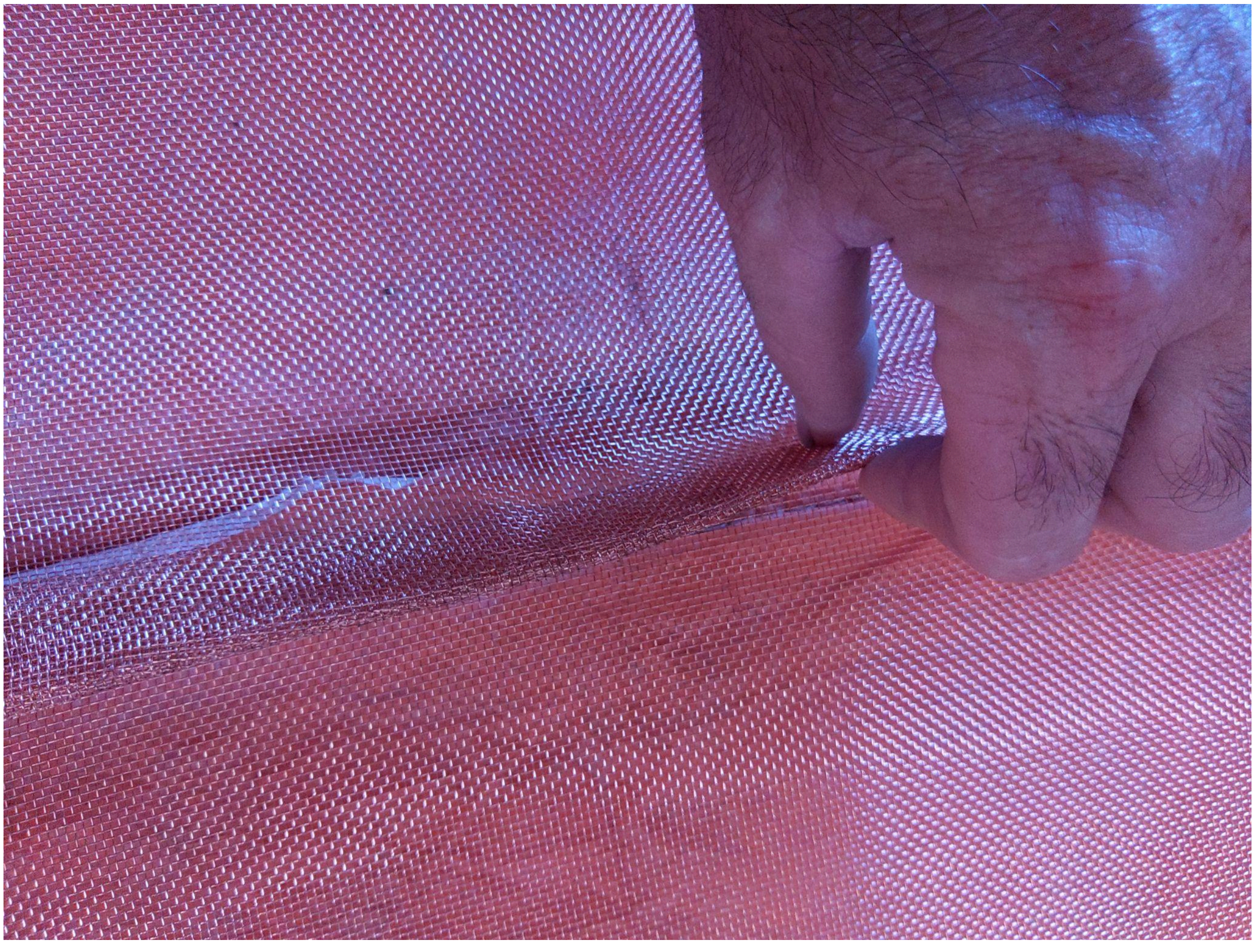
Test 2: The Cellphone Drop (High-Frequency RF Test)

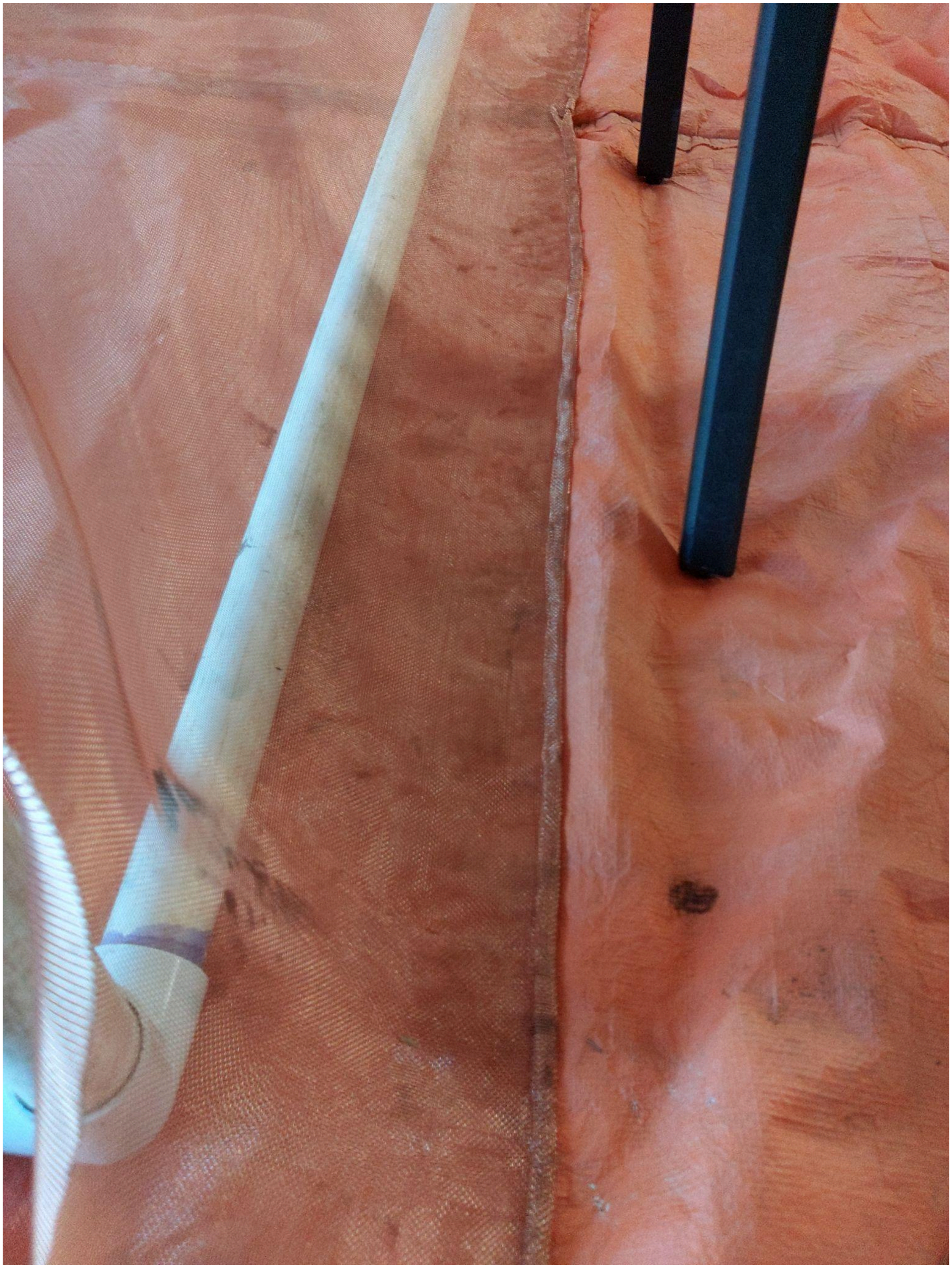
- **Method:** Take your smartphone, disable Wi-Fi calling, and ensure it is connected to the standard cellular network.
- **Execution:** Place it on the plastic chair inside the tent and close the flap. Step outside and dial your phone's number.
- **Success Criteria:** The call must go straight to voicemail. If the phone rings inside the tent, high-frequency microwaves (GHz range) are penetrating your mesh, indicating that your fold-and-tape seams are not crushed entirely flat.

Test 3: The Ultimate Truth (The OpenBCI GUI)

- **Method:** Set up your command station outside the tent (laptop on battery). Run the active USB extension and Fiber Optic HDMI through the floor slit.
- **Execution:** Turn on the OpenBCI Mark IV headset and place it on the chair inside the tent *without a person wearing it*. Seal the flap. Open the OpenBCI GUI on your laptop and activate the FFT (Fast Fourier Transform) widget.
- **Success Criteria:** Look for the 60Hz frequency line. Outside the tent, this should register as a massive spike. Inside the sealed tent, that 60Hz spike should vanish entirely into the baseline noise floor.





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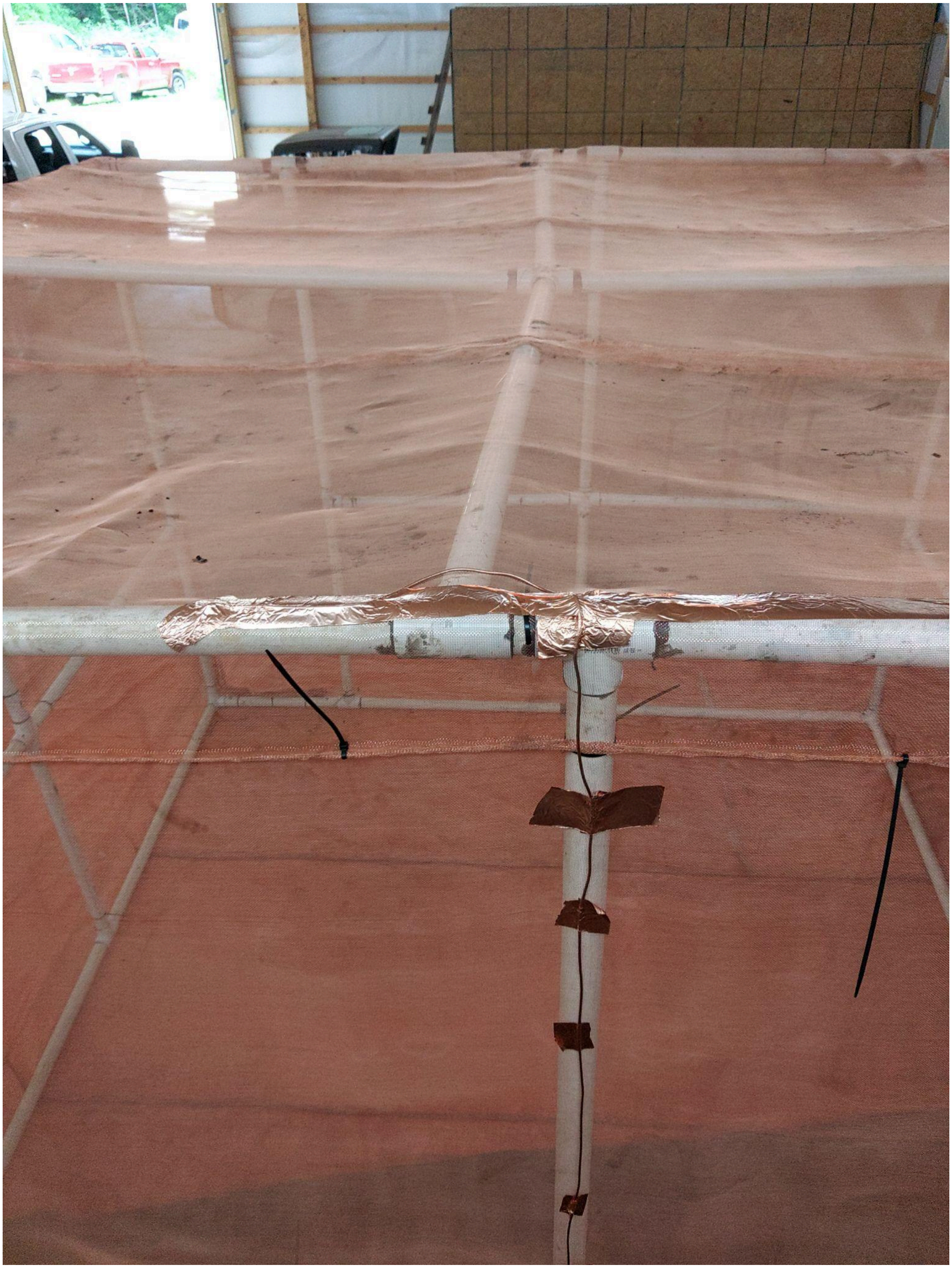

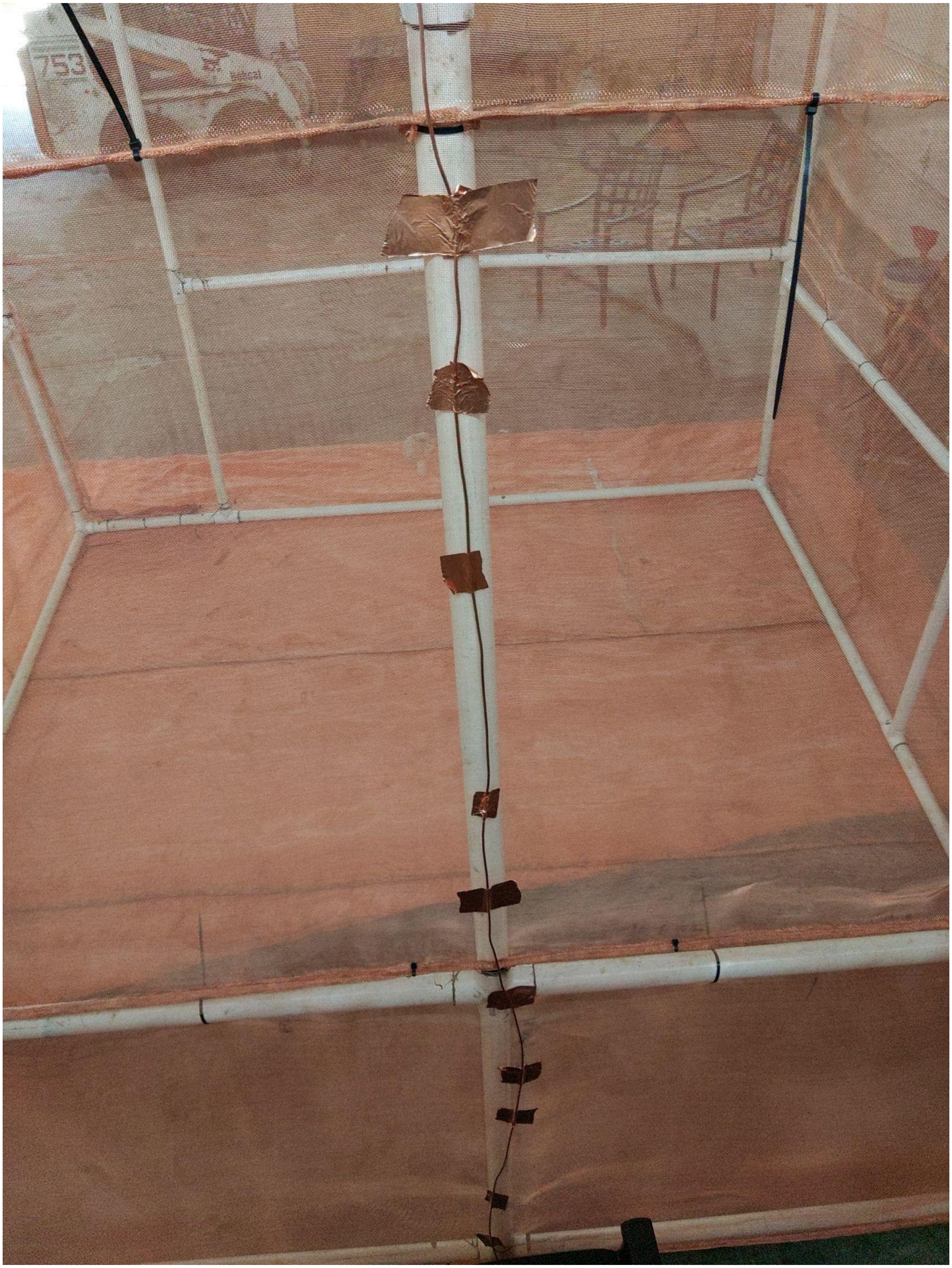


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